EXPERIMENT 19

AIM:

To understand and implement the Logistic Regression algorithm for the given dataset.

CODE:

```
import numpy as np

import pandas as pd


# Load the dataset

df = pd.read_csv('Student_Pass_Logistic.csv')

print(df)


# Display the first five rows

print(df.head())


# Separate features and label

features = df.iloc[:, [0, 1]].values

label = df.iloc[:, 2].values


print(features)

print(label)


# Import required libraries

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LogisticRegression
```

```
# Find suitable random states
for i in range(1, 401):
    x_train, x_test, y_train, y_test = train_test_split(
        features, label, test_size=0.2, random_state=i
    )

    model = LogisticRegression(max_iter=1000)
    model.fit(x_train, y_train)

    train_score = model.score(x_train, y_train)
    test_score = model.score(x_test, y_test)

    if test_score > train_score:
        print("Test {} Train {} Random State {}".format(
            test_score, train_score, i
        ))

# Split the dataset into training and testing sets
x_train, x_test, y_train, y_test = train_test_split(
    features, label, test_size=0.2, random_state=42
)

# Create and train the Logistic Regression model
finalModel = LogisticRegression(max_iter=1000)
```

```
finalModel.fit(x_train, y_train)

# Evaluate the model
print("Training Accuracy:", finalModel.score(x_train, y_train))
print("Testing Accuracy:", finalModel.score(x_test, y_test))

# Generate the classification report
from sklearn.metrics import classification_report

print(classification_report(
    label,
    finalModel.predict(features)
))
```

OUTPUT:

1. Original Dataset

	StudyHours	Attendance	Pass
0	1	50	0
1	2	55	0
2	2	60	0
3	3	58	0
4	3	62	0
5	3	65	0
6	4	60	0
7	4	68	1
8	4	70	1
9	5	65	0
10	5	70	1
11	5	72	1
12	6	68	1
13	6	75	1
14	6	78	1
15	7	72	1
16	7	80	1
17	7	82	1
18	8	78	1
19	8	84	1
20	8	86	1
21	9	82	1
22	9	88	1
23	9	90	1
24	10	86	1
25	10	92	1
26	10	94	1
27	11	90	1
28	11	95	1
29	12	96	1

2. Features

```
array([[ 1, 50],
       [ 2, 55],
       [ 2, 60],
       [ 3, 58],
       [ 3, 62],
       [ 3, 65],
       [ 4, 60],
       [ 4, 68],
       [ 4, 70],
       [ 5, 65],
       [ 5, 70],
       [ 5, 72],
       [ 6, 68],
       [ 6, 75],
       [ 6, 78],
       [ 7, 72],
       [ 7, 80],
       [ 7, 82],
       [ 8, 78],
       [ 8, 84],
       [ 8, 86],
       [ 9, 82],
       [ 9, 88],
       [ 9, 90],
       [10, 86],
       [10, 92],
       [10, 94],
       [11, 90],
       [11, 95],
       [12, 96]])
```

3. Labels

```
array([0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 1,
       1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1])
```

4. Final Logistic Regression Model

```
LogisticRegression(max_iter=1000)
```

5. Training and Testing Accuracy

```
Training Accuracy: 1.0
Testing Accuracy: 1.0
```

6. Classification Report

	precision	recall	f1-score	support
0	1.00	1.00	1.00	8
1	1.00	1.00	1.00	22
accuracy			1.00	30
macro avg	1.00	1.00	1.00	30
weighted avg	1.00	1.00	1.00	30

RESULT:

The Logistic Regression algorithm was successfully implemented and used to classify the given dataset.